

Institute of Actuaries of India

Subject ST2 –Life Insurance

September 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) Appropriation Price can be defined as the price at which the company will create a unit. In other words, it is the amount of money that the company should put into the fund in respect of each unit it creates in order to preserve the interests of the existing unit-holders.

[2]

- ii) This can be calculated as follows:

- the market “offer price” value of the assets held by the fund plus the expenses that would be incurred in the purchase and any stamp or other duty payable in respect of such a purchase
- *plus* the value of any current assets, such as cash on deposit or investments sold but not yet settled
- *less* the value of any current liabilities, such as investments purchased but not yet settled or loans to the fund
- *plus* any accrued income, such as interest income from fixed-interest securities and deposits, net of any outgo, such as fund charges.
- *less* any allowance for accrued tax, if applicable.

This gives the net asset value of the fund on an “offer basis”. Dividing by the number of units existing at the valuation date, *i.e.* before any new units are created, gives the appropriation price.

[4]

[6 Marks]**Solution 2:**

- i) Principles to consider are as below:

- take into account policyholders’ reasonable expectations
- treat both surrendering and continuing policyholders equitably
- at early durations, not appear too low compared with premiums paid, taking into account any projections given at new business stage
- at later durations, be consistent with projected maturity values
- not exceed earned asset shares, in aggregate, over a reasonable time period
- take account of surrender values offered by competitors (and possibly also auction values, where available)
- not be subject to frequent change, unless dictated by financial conditions
- not be subject to significant discontinuities by duration
- not be excessively complicated to calculate, taking into account the computing power available
- be capable of being documented clearly
- avoid selection against the insurer.

[5]

- ii)

The primary reason for consistency is to ensure fairness between continuing policyholder and surrendering policyholder. It would be unfair to treat a surrendering policyholder more favorably than one who is making their policy paid up or vice-versa.

All the marketing implications relating to unfairness therefore apply here.

It also avoids possible problems with anti-selection. For example, if the bases were inconsistent, it might be possible for a policyholder to make their policy paid up, then surrender immediately afterwards, and as a result obtain a better surrender value than if they simply surrendered their original policy in the first place.

This would impact unreasonably on the insurer’s profits.

[3]

[8 Marks]

Solution 3:

i)

a) Mortality Assumption:

Mortality assumption reflects the expected future experience of the lives who will take out the contract being priced.

The expected future experience of the policyholders will depend on the target market for the contract distribution channel involved ; the underwriting controls ; the expected change in the experience since the time of the last historical investigation to the point in time at which the assumption will on average apply.

With respect to the underwriting controls, there is a constraint that the proposal can't be denied on the basis of health grounds. This indicates that the pricing should reflect this constraint and build in appropriate margins for adverse selections and substandard health conditions.

As the insurer was primarily doing unit linked business, it may not have suitable historical data to reflect the expected future experience for a term insurance contract.

The company may have some data to the extent of risk covered in the unit linked products.

However, unit linked products are seen as saving vehicles, and therefore risk of anti-selection is lower. Additionally, the target market of the two products may also differ.

Reliability, relevance and appropriateness of the data will have to be tested before using such data.

The company may also use the standard mortality table and may use appropriate adjustments to it based on the population mortality statistics, industry data or reinsurance data, if any available to the insurer.

Consideration may have to be made to include any expectations of increasing mortality or trends as significant death benefits will be offered under the term insurance contract, and there is no denial of lives through underwriting and the premiums being guaranteed for the term of the policy.

Based on the above, suitable margins may also be built in to accommodate for uncertainty in the data, availability of relevant data, credibility of the available data. [7]

b) Expense assumption:

The insurer incurs expenses in acquisition, processing and administering the insurance contracts initially as well as during the period of the contract.

Commission will be known in advance and hence no assumption needed

The estimate of the expense assumptions needs to reflect the expected expenses to issue and administer an insurance contract

The initial expenses need to allow for initial medical underwriting costs and other cost sales and marketing related costs

Though the company may not be able to deny cover based on underwriting, they would want to classify the lives well so that the right terms can be quoted.

Therefore, the underwriting costs may be high.

Also need to consider claim/termination expenses. Given this is a term assurance the investment expenses are likely to be minimal, but an assumption may be included

The expense assumption may also include a contribution to the company's fixed overheads.

The expense assumption may consider the recent experience of the company for the existing product, and consider how this may translate for the term product.

For example, the overhead costs may not change, but underwriting, claims handling costs and investment expenses would change.

The assumption would need to consider projected budgets for the sales staff and the business volumes.

May need to use external data from reinsurers, or industry statistics

The policy has a long term - hence the estimation of expense should also consider the expected future rates of inflation.

Inflation may be based on future expectations of wage inflation in the economy, or other source of information

Also appropriate margins to be included in the estimation to reduce adverse future experience.

Especially given this a new product for the company

[6]

ii)

- Reinsurance
 - The company has so far sold only unit linked business, which usually has low sums at risk.
 - With term insurance, the sums at risk on mortality are likely to be higher, so the company would need to review its reinsurance program.
- Underwriting
 - Since unit linked products are usually savings products, risk of anti-selection, and the role of underwriting, is lower.
 - For term insurance more stringent mechanisms of underwriting may be required
 - Additionally, clear guidelines would need to be set on the terms that are to be offered to substandard lives since the regulation disallows rejection
- Claims processing
 - To avoid anti-selection this might need to be significantly enhanced.
- Experience Analysis
 - Given the critical role mortality assumption plays in the pricing and reserving of term insurance products, the mortality experience analysis may need to be strengthened / reviewed / done more frequently.

Award marks for any other valid risk management practice / process that is mentioned by the candidate, and would be impacted by the introduction of the term insurance plan.

[5]

[18 Marks]

Solution 4:

i)

1. It provides a saving mechanism for the insured.
2. Being unit linked, it is expected to provide higher returns for the perceived higher risk.
3. With the feature of linking the death and maturity benefits to the guaranteed price, the customer can enjoy the downside protection on both death as well as on maturity in a unit linked product
4. ...And still expect the overall benefits to provide protection against the inflation since the investment is in a market-linked fund.
5. The product provides a small death benefit (linked to the guaranteed price) hence provides customer with some protection for dependents

[3]

ii)

1) Risks to the Insurer:

1. The product feature would rely heavily on systems and models. This could introduce model / system risk.
2. This may be true for any unit linked fund, however, due to the large volume of funds that are involved in a single product the complications would just multiply for these types of products.
3. Too many small sized close ended funds, in the long run, which would reduce in size due to exits like surrenders, deaths and other benefits
4. and eventually may not be economical to generate sufficient income.
5. and be costly to maintain
6. As the fund size reduces, the funds may not provide any real returns to the policyholder, unless these are merged with other funds — which would again create an issue of the applicable guarantee.
7. There would be a lot of administrative work and operational issues involved in managing the huge number of close ended funds
8. which could eventually create systemic / operational risks within these structures.
9. As insurer has the flexibility to invest in any asset category, there is a possibility that the insurer may manage the fund at a fairly low risk such that the guarantee risk remains low at all points of time.
10. To that extent the product may lose its attractiveness in terms of yielding higher returns.
11. This is particularly true if there is a market crash and the guaranteed price is then fixed for the rest of the policy and considerably higher than current price
12. Which will lead insurer to adopt an even more cautious strategy and this may go against policyholder expectations
13. Insurer is subjected to sudden changes in the value of investments possibly leading to default risk, operational risk and liquidity risk.
14. If guarantee price is higher than current price then the insurer will need to set aside additional reserves
15. Which may lead to solvency issues
16. Due to this complex procedure being involved, understanding the entire process for the policyholder may be a concern and can lead to confusion, misleading and mis-selling.
17. Which can in-turn cause reputational risk
18. There is risk that the guarantee is not priced appropriately
19. There is risk that the charges collected from the product are not sufficient to meet the expenses incurred under the product
20. ... both these risks may reduce if the charges are reviewable under the product
21. Though the extent to which the charges may be restricted by regulations, TCF and competition
22. Risk of high surrenders
23. ... although the guarantee does not apply at the time of surrender, high surrenders would reduce the number of policies in-force and increase the per policy expenses
24. Exposure to mortality risk, particularly since the guarantee will usually bite throughout the term of the product.

2) Risks to the Insured:

1. The product will provide a very low death benefit in the early years
2. And the sum assured on death or maturity is not certain
3. The policyholder bears all the investment risk

4. Although some of this is mitigated by the operation of a guaranteed price
5. The products are more complex and confusing even to a financially literate person.
6. Scope for mis-leading advertisements and illustrations.
7. Each premium paid by the policyholder will go to a different fund and the highest guarantee NAV applicable to each premium could be different.
8. The policyholder has to track each fund into which his premium is paid, to enable him to know the entire fund value under the contract at any point of time.
9. In most cases, the policyholder may not even know in which fund his premium is allotted. In this process, the entire transparency is lost to the policyholder.
10. In the long run the benefits that these funds would generate may not be much different from the other ULIP funds,
11. due to inherent management of the guarantee.
12. There may be an additional management charges for these funds, increasing the cost to the policyholder
13. Cost of guarantee – Insurer may levy a charge for the guarantee risk, which may not be understood by the policyholder.
14. The policyholders may have a palpable feeling for upward NAV growth, which may not be true.
15. The returns and effectiveness of the products depends entirely on the efficient management of the underlying fund.
16. There is a scope that the company may deliberately manage the NAV.
17. As the company has control over the investment policy, this could result into conflict between investing to meet the guarantees and investing for maximum performance.
18. This may go against the reasonable expectations of policyholders.
19. Risk that the policyholder needs to withdraw the fund in emergency and then loses the guarantee
20. There is the risk of insurer insolvency

[16]

[19 Marks]

Solution 5:

i)

1. Reinsurer may want to increase their market share.
2. The existing rates may have been based on their experience from other markets, since they might not have had adequate experience within this geography
3. They might have analysed the experience for this block of business and found that their current rates were too high
4. There might have been a change in the mortality experience due to mortality improvement since the time the reinsurance rates were last reviewed
5. They might have been facing significant competitive pressure
6. They might wish to encourage online term insurance business over the business from other distribution channels because of differences in experience they might have observed, e.g. online channel might have higher sums insured leading to higher reinsured percentage.
7. Leading reinsurers have more capacity to assume larger blocks, and capacity – coupled with strong balance sheets – also allows more risk taking, and potentially higher profitability.
8. Larger, more diversified reinsurers enjoy superior diversification benefits and may have greater options to optimize their own balance sheets, particularly with retrocession placements and risk transfer to capital market

9. Greater diversification allows greater ability to rebalance the portfolio by dynamically increasing or decreasing the size across products and markets, as local conditions continuously change. [5]

ii)

1. Term insurance products are very price sensitive
2. Online platform, makes comparison even easier, and making them even more price sensitive.
3. The intensity of impact on the industry will depend on the level of reinsurance prevalent in the market
4. If the reinsurance rates been reduced for existing business as well, it will lead to profits for the insurance companies that were reinsuring business with this reinsurer
5. Increased profits in the market may lead to new entrants
6. Other companies may pressurise their reinsurers also to reduce the rates
7. Or they may migrate to this reinsurer, increasing its market share
8. For new business, the insurer(s) may choose to pass on some (or all) of this profit to the insured in an attempt to increase market share
9. This may lead to over all price wars, given the highly competitive market, and lowering of term insurance premiums for online business
10. The gap between the online term insurance prices and other distribution channels may increase
11. ...assuming that the online product is cheaper in view of lower expenses.
12. ... leading to increase in online distribution channel sales
13. Lowered premium rate may also increase the term insurance coverage in the market
14. Due to increased online sales, the need for higher sum assured policies may be increased.
15. If the underwriting capacity of the insurer is limited, there could be a scope to reinsure more depending on the availability of the reinsurers, thereby increasing its market share. [6]

iii)

1. If insurance companies move greater risk to the lead reinsurer in order to take advantage of the reduction in prices there might be concentration of risk with one reinsurer,
2. causing higher counter-party risk for the market.
3. In case the company already reinsures with the lead reinsurer, there could be a possibility of increase in counterparty default risk since the reinsurer has reduced their reinsurance rates.
4. The reinsurance rates offered may not be guaranteed for the term of the contract for the existing block of business, thereby possibly creating a pricing risk to the insurer.
5. Reinsurance rates may be level over the life of the policy or increasing every year based on attained age. Depending on the type of premium rates offered and the corresponding guarantee offered by the insurer, the risks to the insurers will change.
6. Due to competitive pricing, the risk margins and the profits expected may decline
7. and the relative importance of expenses assumptions may increase, depending on number the policies inforce.
8. The insurer also needs to be conscious of the potential for “channel conflict”. The traditional distribution channels may raise concerns that lower pricing of online term product could hurt their business.
9. The reinsurance treaty conditions may not be exactly same as that of the online policy terms and conditions, creating a gap in the terms of offering of the online contract.
10. This may increase the insurers risk of not being able to meet its obligations.

11. There could be additional administration required given the insurer will need to identify different sets of reinsurance rates and which policies are covered.
12. which in turn increases risk.

[8]

[19 Marks]**Solution 6:****i) List assumptions**

The Following are the assumptions that needs to be considered:

- a. Mortality / longevity assumption
- b. Mortality improvement factors
- c. Maintenance Expenses
- d. Investment Expenses
- e. Expense inflation rate
- f. Investment return
- g. Discount rate
- Taxation (if applicable)

[3]

ii) Factors in determining BE assumptions

Best estimate liability (BEL) will be generally used when the management wishes to have best estimate of the company's financial performance.

Generally will set assumptions based on past experience investigations

And will probably consider alongside pricing assumptions initially

In setting the above assumptions care needs to be taken to include no prudence in the assumptions.

Mortality assumptions may be derived based on recent experience of the company, if available, else pricing basis may be used as a starting point.

Since longevity is a major challenge in annuity contracts, allowance needs to be made to allow for possible improvement in mortality in future. These can be done by incorporating improvement factors to the mortality assumption assumed.

The mortality improvement factors may be based on company's own experience by looking at cohort of policies of similar age and then their mortality improvement over the time period.

If company own experience is not available then census data may be used.

Or use data from reinsurers or industry

Future expenses to be assumed in calculation of liability will be based on company own realistic estimate of likely future expenses in management of the annuity portfolio.

Taking results of expense investigation

Investment expenses may be assumed to be a reduction in investment return

Since annuity contracts are long term in nature, assumption regarding future expense inflation is critical.

These may be derived based on the national statistics like Consumer Price Index, WPI, etc.

Investment return assumption would be based on expected returns on assets backing the annuities

Which are likely to be combination of government and corporate fixed interest

And discount rate will be based on shareholder expected rate of return

Regulations may prescribe that risk free rates are used for both investment return and discount rate
And hence the assumption will depend on the purpose of the valuation

Need to allow for any taxation based on current tax regime

Would need to comply with any regulations if the liability is being published [8]

iii) Process for calculation of best estimate liability using cashflow method may be as follows:

First a cashflow model needs to be built which will project all the possible cashflows arising

... may use a bespoke actuarial software package

...for example annuity benefit outgo, expenses,.

... investment return, tax outgo etc

Company may use an individual policy level model

or use model points

of homogeneous policies

split by appropriate risk factors

If grouping of model point is done then this needs to be tested first to ensure that there are no material impact due to grouping.

various cashflow then need to be projected into future

using assumptions as set above

and discounted back at discount rate [5]

iv) Risk Margin will cover those elements of the basis that cannot be hedged

Such as mortality and expenses

Starting point will be best estimate assumptions

Need to add a risk margin for inherent uncertainty in these assumptions

Can be calculated as a margin added into each assumption

Or calculate using cost of capital approach

Steps involved in cost of capital approach are as follows:

1. First identify the various risks underlying the product and which has not been hedged.
These can be the longevity risk and expense risk
2. Then project forward the future capital that the company is required to hold in respect of these risks, where this is determined at the end of each projection period (*eg* year) during the run-off of the business.
3. The projected capital is determined according to the relevant regulatory basis,
4. and is the amount required to be held in excess of the projected liabilities at each period.
5. The projected capital amount is then multiplied by the cost of capital rate.
6. Cost of capital rate represents the frictional cost of not able to invest in higher return generating assets
7. or the incremental cost of raising capital over and above the risk-free rate.

8. The projected capital amount multiplied by the cost of capital rate is then discounted to valuation period using the market consistent discount rate i.e. risk-free rate to arrive at the risk margin.
9. It is important to bear in mind that the projected capital should be as per the regulatory regime of the country.
10. If the local regulation requires Risk Based Capital then the calculation of risk margin needs to appropriately allow for the correlation of various risks and their interactions.
11. The Company may also simplified method like run-off driver approach to calculate the risk margin.

[8]

- v) In theory, a market-consistent value of a liability is the price that someone would charge for taking responsibility for (*i.e.* ownership of) the liability, in a market in which such liabilities are freely traded.

In the usual absence of such a market, an approximate approach has to be taken.

Most of the best estimate assumptions will be appropriate for a MC valuation
 However investment return will need to be based on risk free rate
 As will discount rate

Risk-free rates may be determined based on government bond yields or on swap rates.
 Appropriate deduction for credit risk needs to be made to the risk-free rate.
 Swap rates will be used if there is sufficiently deep and liquid market in the country.

If the company has invested the assets backing the annuity liability in corporate bonds, the actuary may want to set the discount rate from corporate bonds rather than government bonds.

This may be done if it is so permitted by local regulations.

Corporate bonds generally carry higher risk than government bonds and hence have higher yields. The actuary may take credit of this higher yield, called illiquidity premium, while setting the discount rate assumption.

Credit for illiquidity premium can only be taken if the long-term liability is predictable in nature, if the insurer has held corporate bonds by matching the duration of the liability and where these bonds are held to maturity.

Since the insurer is not exposed to the risk of changing spread on such assets, although appropriate deduction for credit risk needs to be made.

[6]

[30 Marks]
